What we did

RL on #230's <contacts-v1.multi> checkpoint, with the reward computed over the **sections of a single rollout** rather than over the rollouts of a group.

Six reward designs, seven runs, **37 scored checkpoints**, all on 8 × A100 via SkyRL: **M-C** each section's contribution to its own rollout's consensus · **M-F** the last section's F1 · **M-B** the best section's F1 (oracle) · **M-BC** best + consensus · **M-FC** last + consensus · **M-K** the rollout's own consensus R-precision · **M-0** a zero-LR control.

Why

#208 ran eleven runs across five rewards and none improved consensus R-precision. The mechanism was a **unit mismatch**: the reward scored one rollout, while the metric votes over 100 *independent* rollouts — an object no rollout can see.

Under <contacts-v1.multi> one rollout emits ~22 contact sets in a single generation, so a reward on those is computed on the kind of object the metric scores, with credit assignment inside the sequence.

The result

Best consensus 0.5806 (M-K), best oracle 0.5663 (M-B), best final section 0.5267 (M-C) — three criteria, three different arms. M-K is the only arm to improve **all four** aggregation modes with every CI excluding zero, on every cut.

The primary criterion is not met. 22 independent plain rollouts read 0.5896; M-K is 0.0090 short. At a larger budget the two draw level — M-K pooled 0.6098 against plain-100's 0.6058, a paired CI that includes zero. So RL **closed the gap between the multi format and ordinary sampling; it did not clearly open one.**

#208's negative result is a dose-response, not a verdict

Consensus against distance moved: 0.5673 at KL 0, **0.5806 at 0.016**, 0.5529 at 0.031, 0.3969 at 0.486. Every reward helps at small KL and damages at large.

#208 ran its arms at KL 0.06–0.10 and to 3.96 — past the peak on all of them — and its two arms under 0.0015 never moved. **The window it needed lay between the two learning rates it tried.** Two runs at a 3.3× different rate agree to 0.002 at matched KL, so this is distance, not schedule.

Reward design, in three parts

E[r] = 0 is necessary and not sufficient. M-C's per-section marginal is **366× larger at 1 section than at 22** — it paid the policy to emit a worse answer — while E[A] = 0 held exactly, because centring is computed *inside* the quantity being gamed.

Every arm moved its candidate count in the direction its own reward pointed. M-C's gradient is strongly negative (collapsed to 1.1 sections); M-B's strongly positive and self-paying (grew to ~26 and held); M-F's nearly flat — and **a weak gradient is not a safe one, it is an unconstrained one**: M-F ran to 259 sections carrying 1.4 contacts each.

Gates written against the failure you have seen will miss the next one. All three original criteria are one-sided; M-F's failure pushed every one *away* from its threshold and none fired.

The arm that won was designed from a failure

M-C's collapse was traced to a scale bug: its per-section marginal is 366× larger at one section than at 22, so it paid the policy to emit fewer candidates. The diagnosis named the repair — a **rollout-level** base that *falls* when sections are dropped.

M-K is that base with nothing else on top: reward = the rollout's own consensus R-precision, GRPO-centred. Section count then held flat at ~22 for 48 steps, and it produced the experiment's best consensus. It is also the only arm neither killed by a gate nor diverged — it ran out of scheduled steps.

Within-sequence credit assignment was the hypothesis's mechanism, and it turned out not to be needed. What was needed was for the reward to be computable on the object the metric scores — which is what the multi format makes possible.

Two results that outlive the experiment

Selection is dominated by aggregation. An ORACLE selector of one draft reads 0.5646 where simply voting the same drafts reads 0.5750. A final section should synthesise its predecessors, not pick among them.

The spread is the resource. Making candidates more uniform raises mean section F1 from 0.432 to 0.532 and *lowers* best-of-22 — the same trade that defeats every sharpening reward here, appearing in the corpus's own size law.

What next

Train M-K further. It is the only arm that stopped for its schedule rather than for a reason. Every other arm answers "how long can you train?" with "not long"; M-K has not been asked.

Then turn on the shaping and blend terms. M-K is the beta = lam = 0 corner of the derived arm. The base alone beat every shaped and blended arm, so shaping is now a hypothesis to test, not a fix to apply.

Nothing further on M-B at lr 1e-5. Two rates, 10 checkpoints, a smooth peak, agreement to 0.002.

The diversity gap is a corpus question. One rollout's sections cover 658 distinct pairs against 1,065 for 22 independent rollouts.

Figures

curves_accuracy.png — all 37 scored checkpoints against training step.

curves_reward.png — each arm against its own reward.

dose_response.png — R-precision against distance moved.

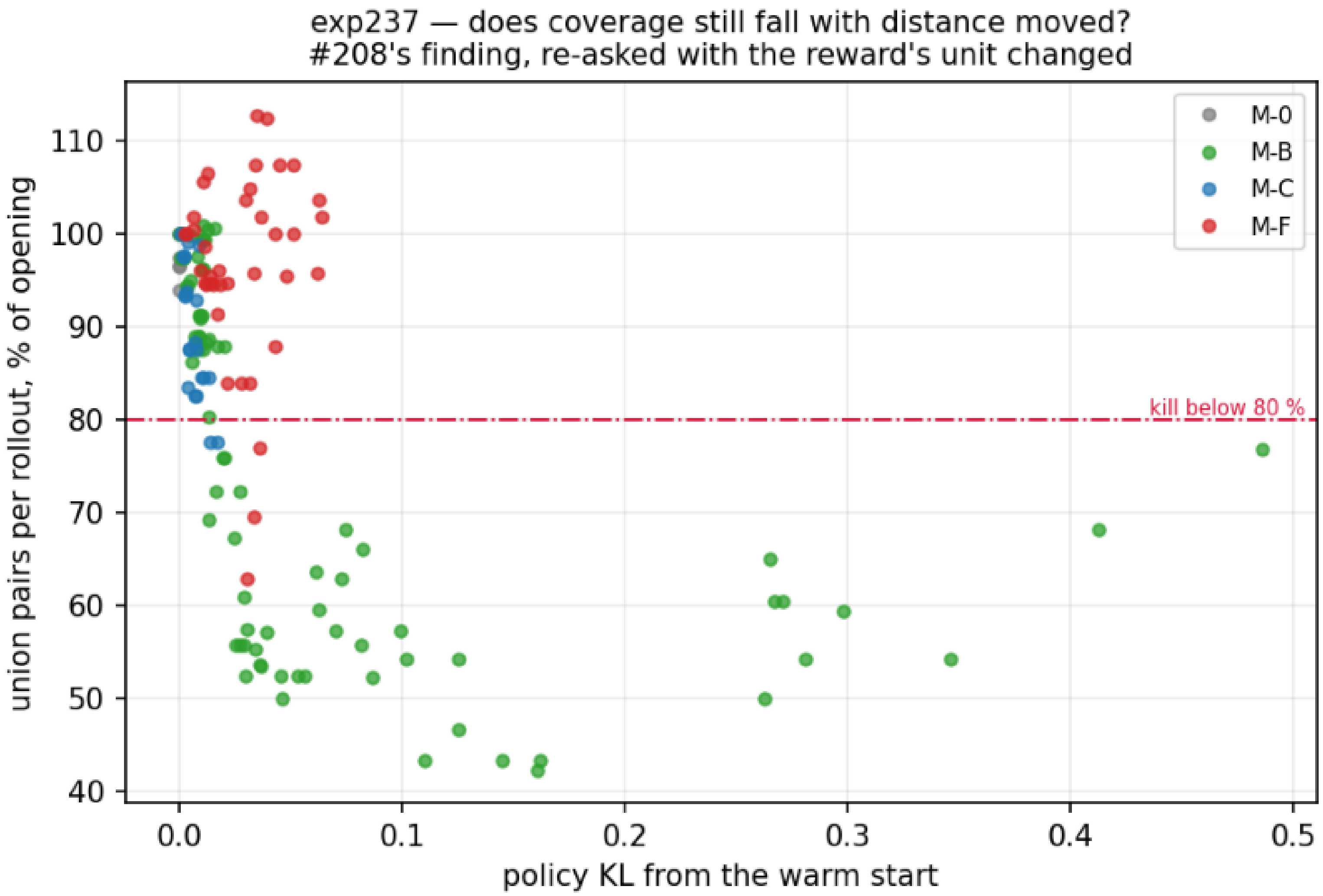
reward_vs_section_count.png — each reward's gradient in the count direction.

section_count_incentive.png — M-C's scale pathology, 366× at one section.

gates_over_training.png · reward_shape.png · coverage_vs_kl.png

coverage_vs_kl.png

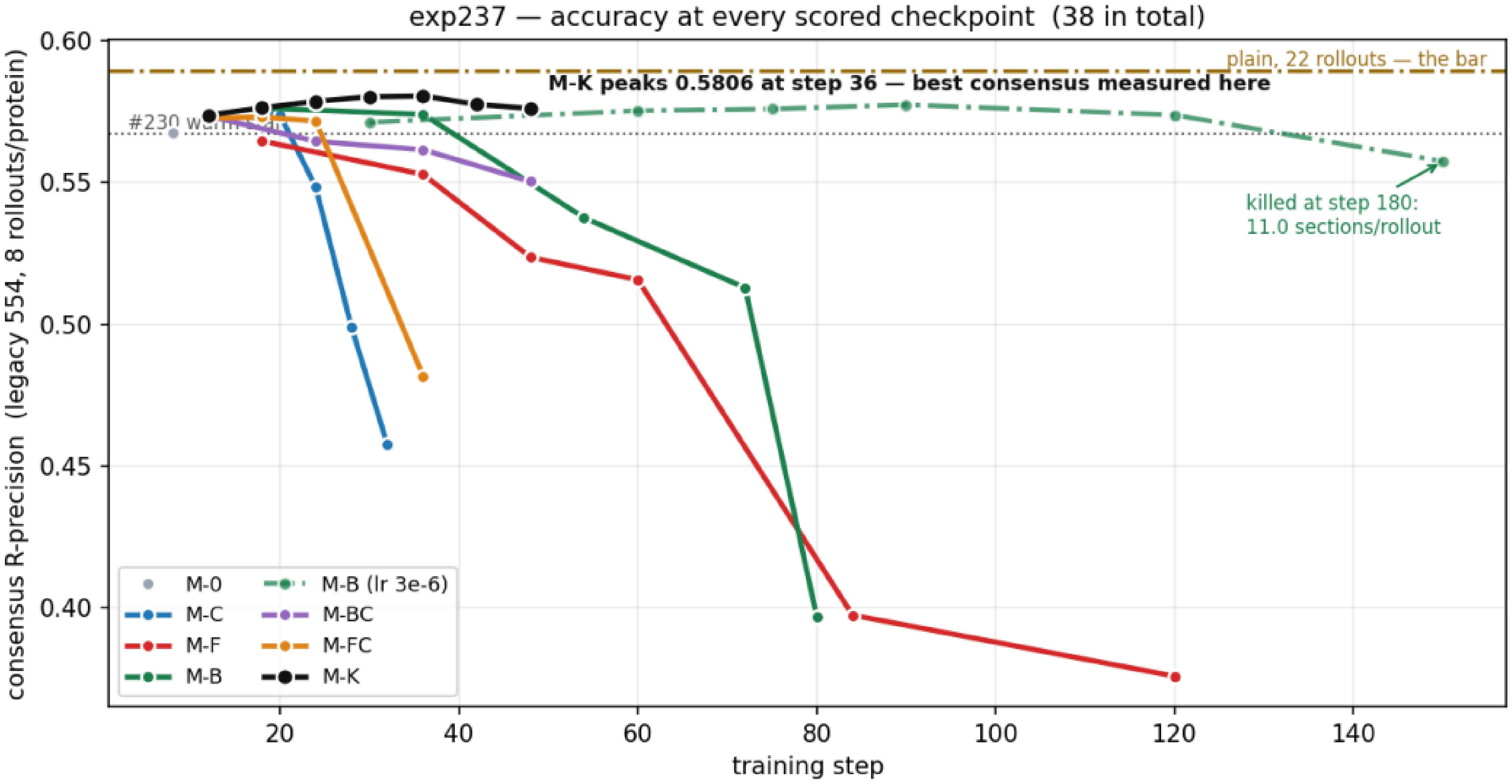
Union coverage against distance moved. M-C crosses the 80% floor by KL 0.013, M-B by 0.016, M-F not until 0.036 -- same destination, different cost per unit of KL.



`$ make_plots.py --steps data/training_steps.csv.gz --out plots/`

curves_accuracy.png

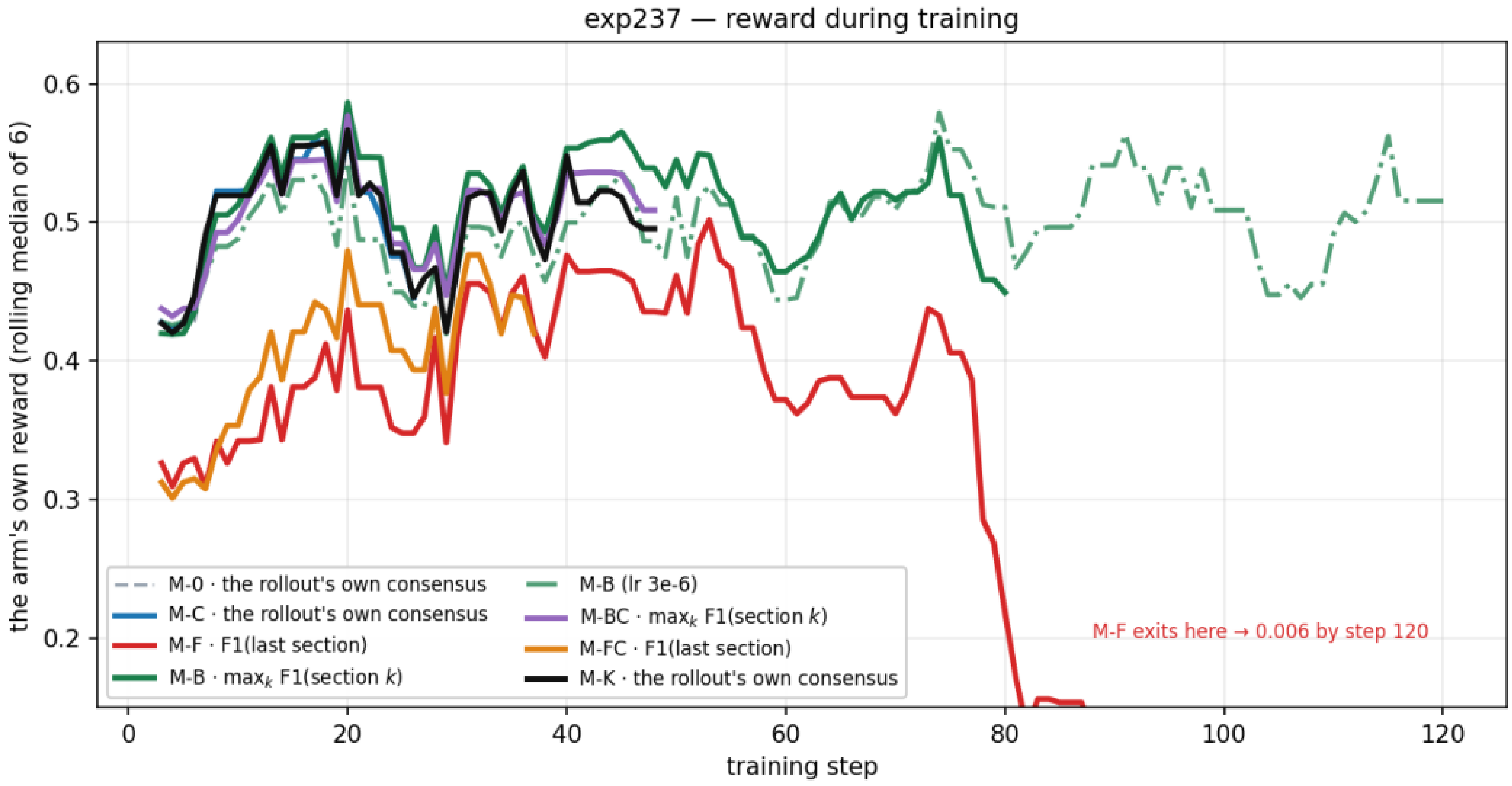
Consensus R-precision on the 554-protein exp89 benchmark at all 37 scored checkpoints. Five arms peak by step ~20 and fall away; M-K peaks later, at step 30-36, and highest.



$ plot_reward_and_accuracy.py

curves_reward.png

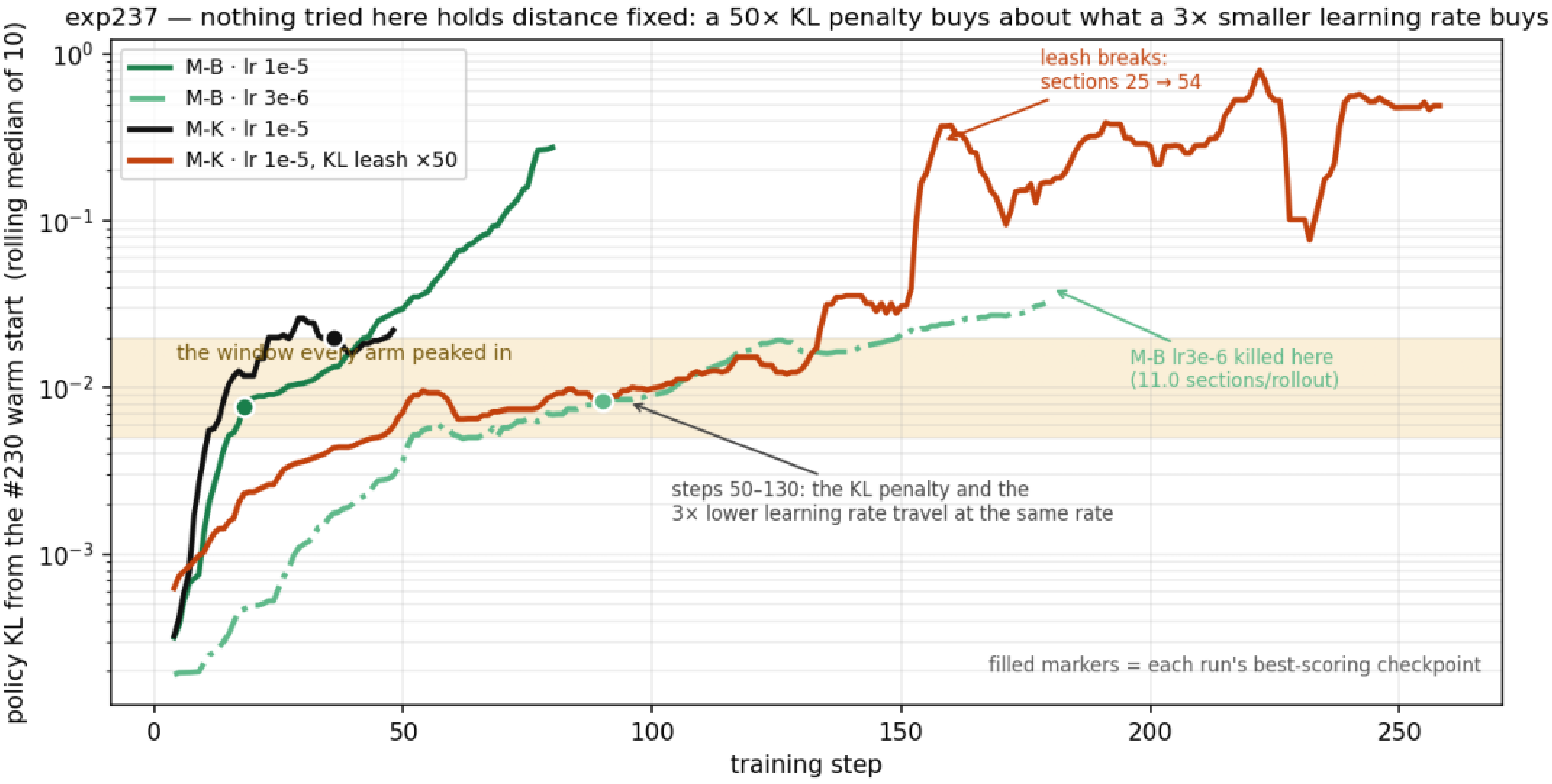
Each arm against its OWN reward — the arms do not share one. Rolling median of 6 training batches; the raw per-batch series is dominated by the protein draw.



$ plot_reward_and_accuracy.py

distance_vs_steps.png

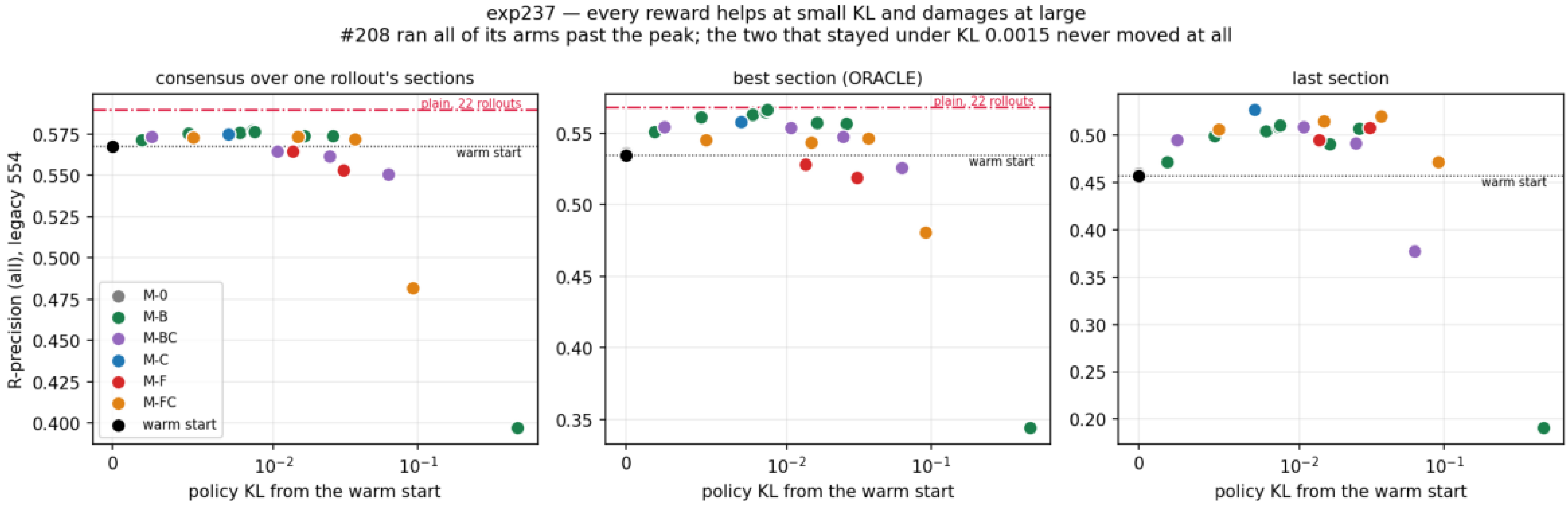
Distance moved against steps taken. A 3.3x smaller learning rate takes ~5x the steps through the same KL window and then fails the same way. Raising kl_loss_coef 50x does NOT hold distance fixed — from step 50 to 130 it tracks the lr-3e-6 run, then runs away to KL 0.5.



$ plot_distance_vs_steps.py

dose_response.png

R-precision against distance moved. Consensus peaks at KL 0.007-0.016; #208 ran every arm past it. Red = the budget-matched bar, which nothing reaches.

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$ plot_reward_shape.py --data data --out plots
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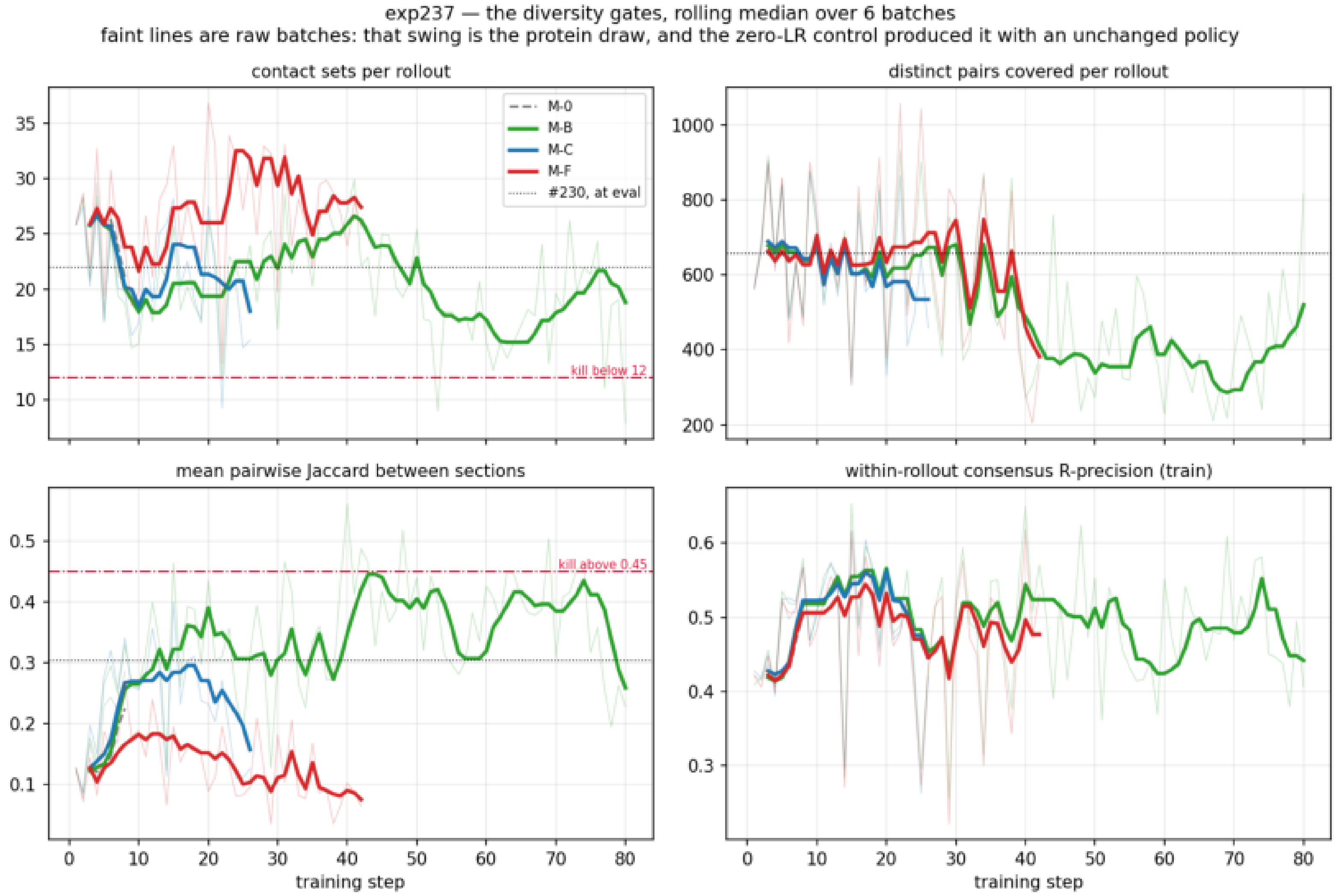
gates_over_training.png

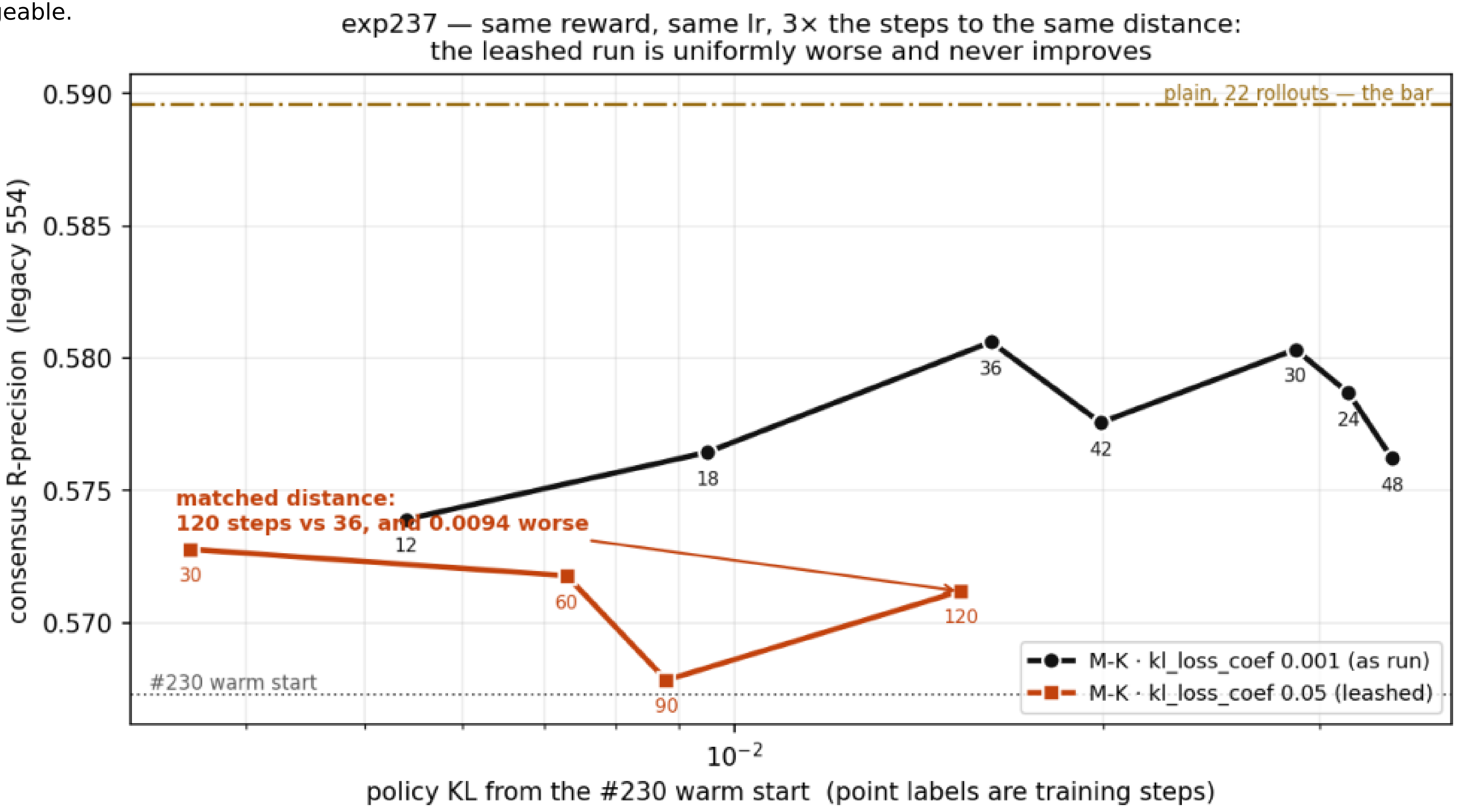
The diversity gates per batch, rolling median over 6. Jaccard separates the arms: M-B rises, M-C and M-F fall. Faint = raw batches, i.e. the protein draw.



`$ make_plots.py --steps data/training_steps.csv.gz --out plots/`

leash_vs_free.png

Arm M-K with and without a 50x KL penalty, plotted against distance moved so the extra steps are a vertical comparison at fixed x. The leashed trace is below the unleashed one at every matched distance and is flat across steps 30-120 — the path to a distance is not interchangeable.

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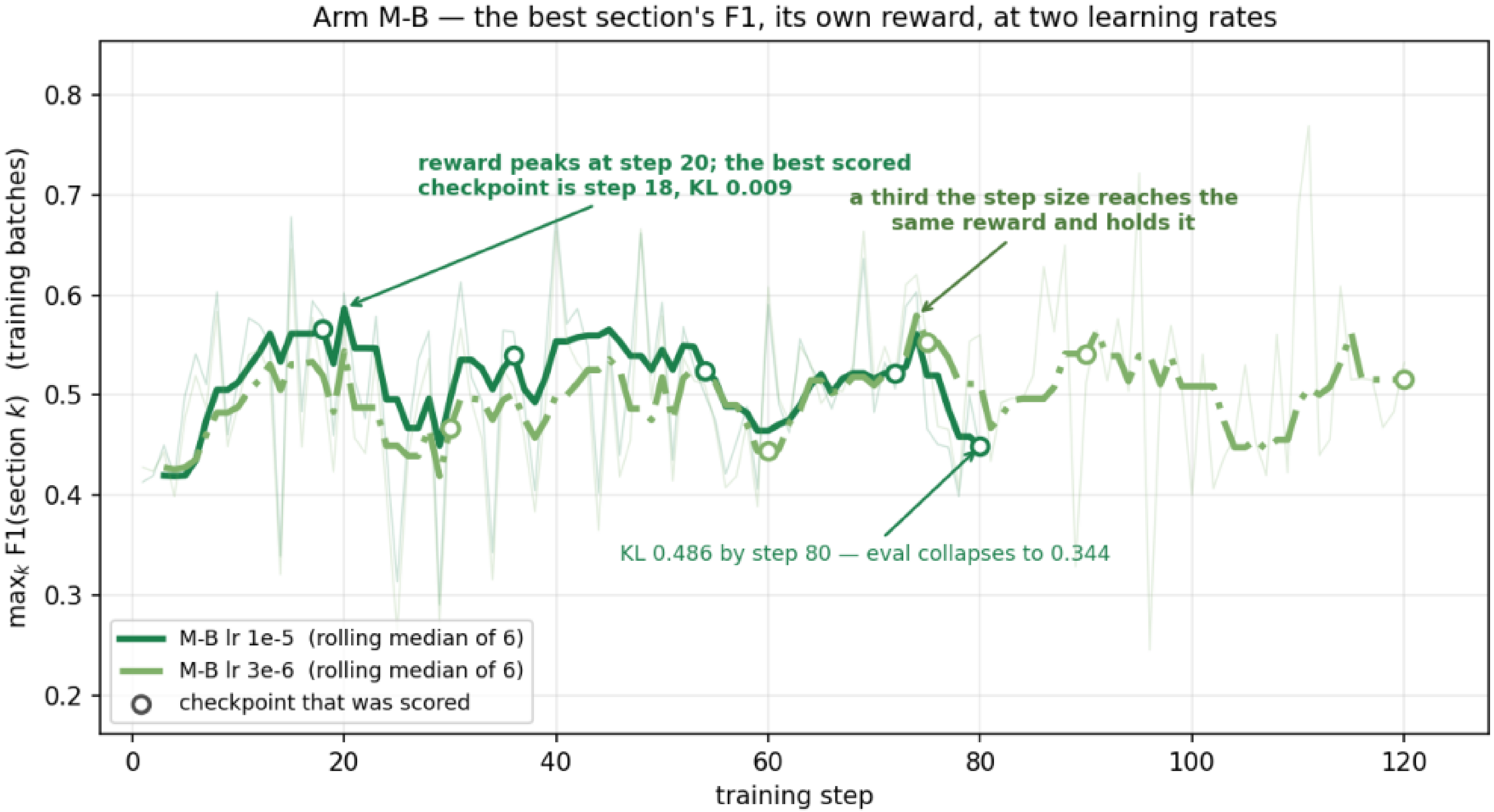
$ plot_leash_vs_free.py
```

reward_curve_m_b.png

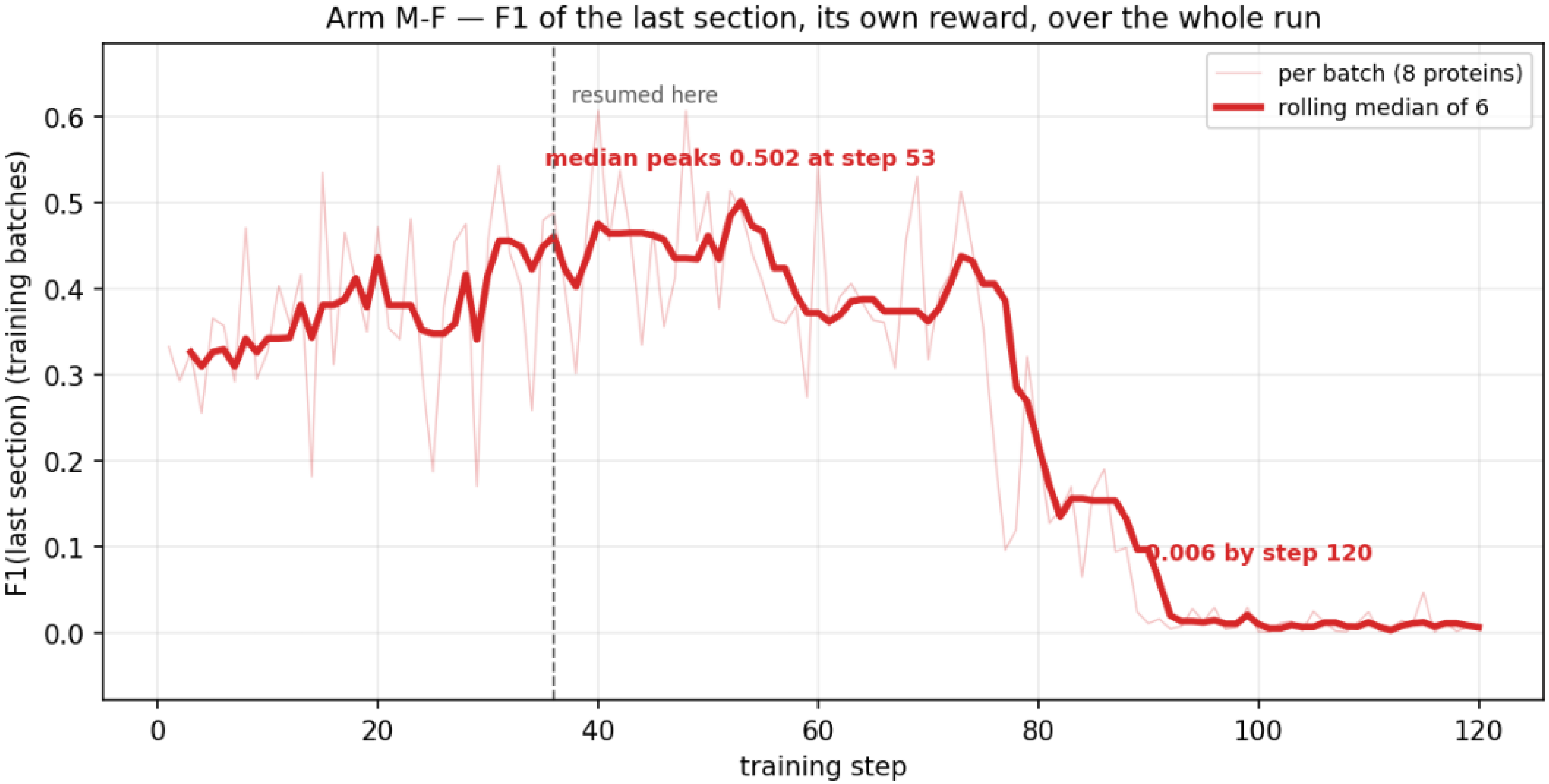
M-B's own reward at lr 1e-5 and 3e-6. Both peak near 0.58; the slow walk holds it while the fast one runs to KL 0.486 and its eval collapses to 0.344. Open circles mark the checkpoints that were scored.



$ -

reward_curve_m_f.png

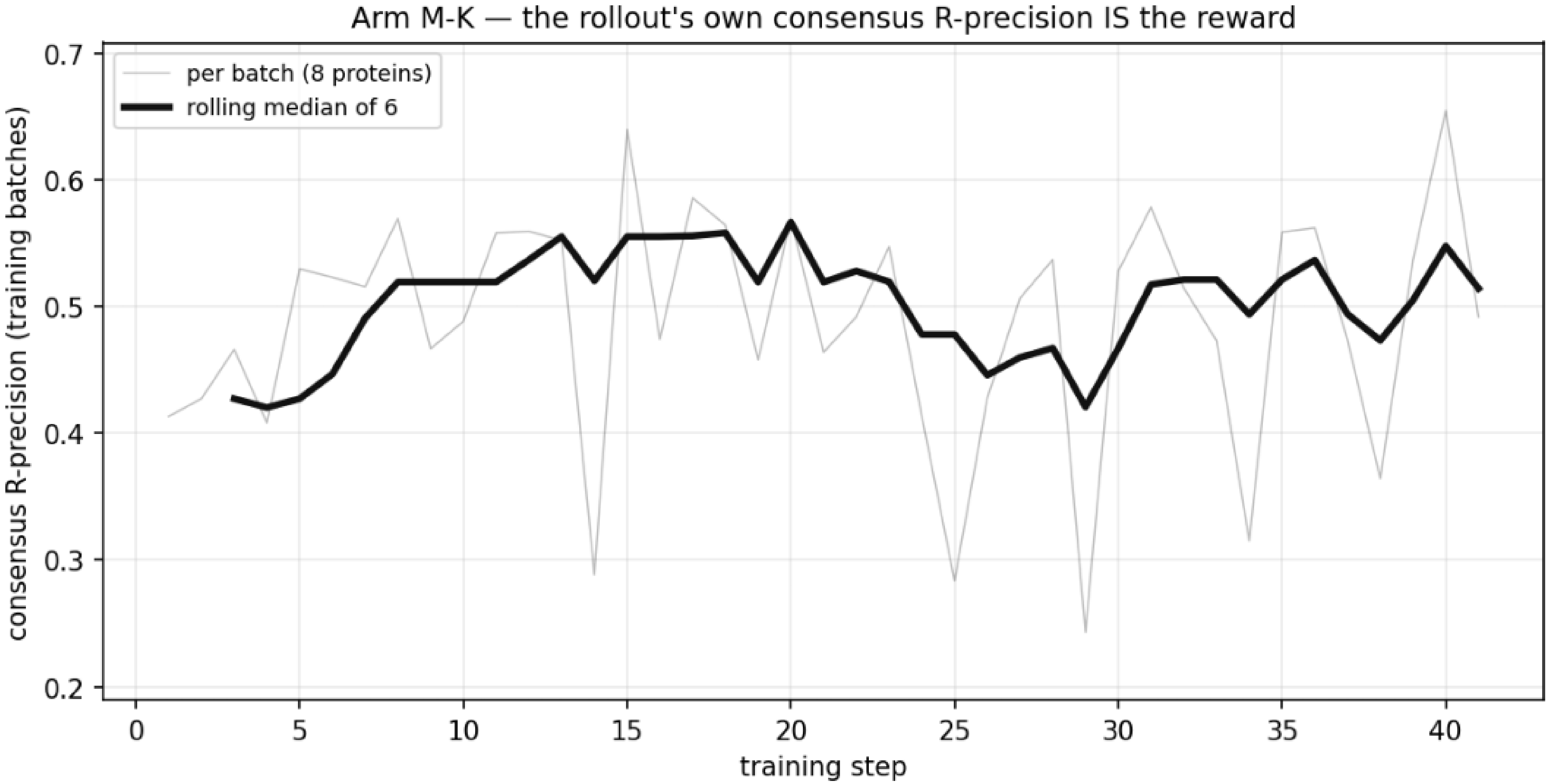
M-F's reward over both runs. It peaks at step 48 and is at 0.006 by step 120: the policy found the unconstrained direction (259 sections of 1.4 contacts each) and took the reward down with it.



$ single.py

reward_curve_m_k.png

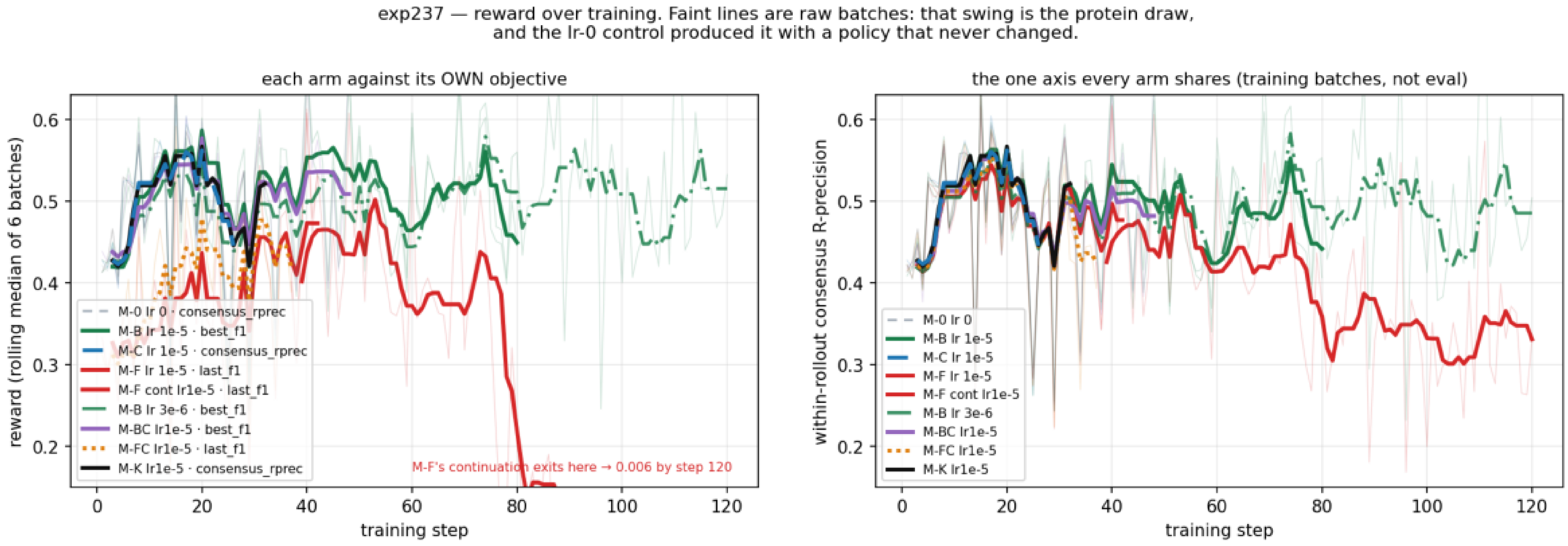
M-K's reward is exactly the quantity plotted: the rollout's own consensus R-precision, one scalar per rollout with a GRPO baseline.



$ single.py

reward_curves.png

Each arm against its own reward (left) and the shared consensus axis (right), rolling median of 6 batches. Training-batch values on 8 proteins, not eval.

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$ plot_reward_curves.py --csv data/training_steps.csv.gz --run M-F cont lr1e-5=/tmp/claude-1000/logs_final/exp237_m_f_lr1e-5.log --offset M-F cont lr1e-5=36 --run M-B lr3e-6=/tmp/claude-100
```
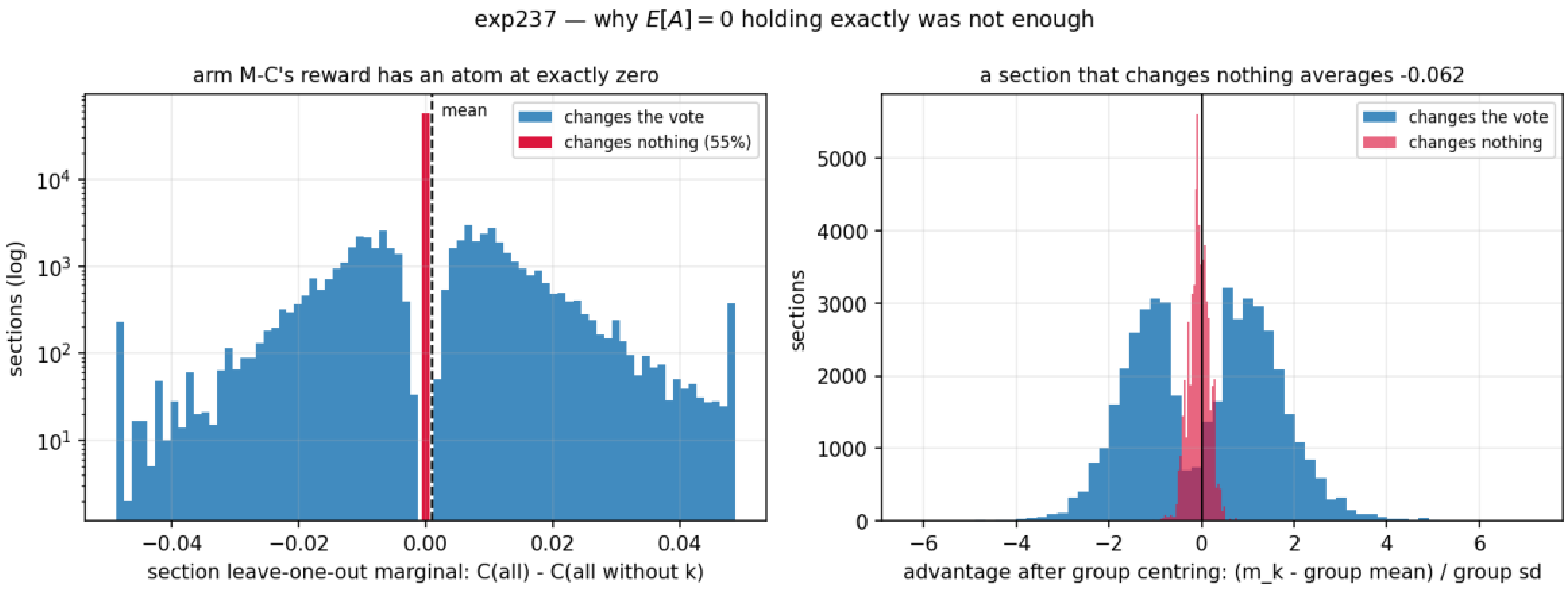

reward_shape.png

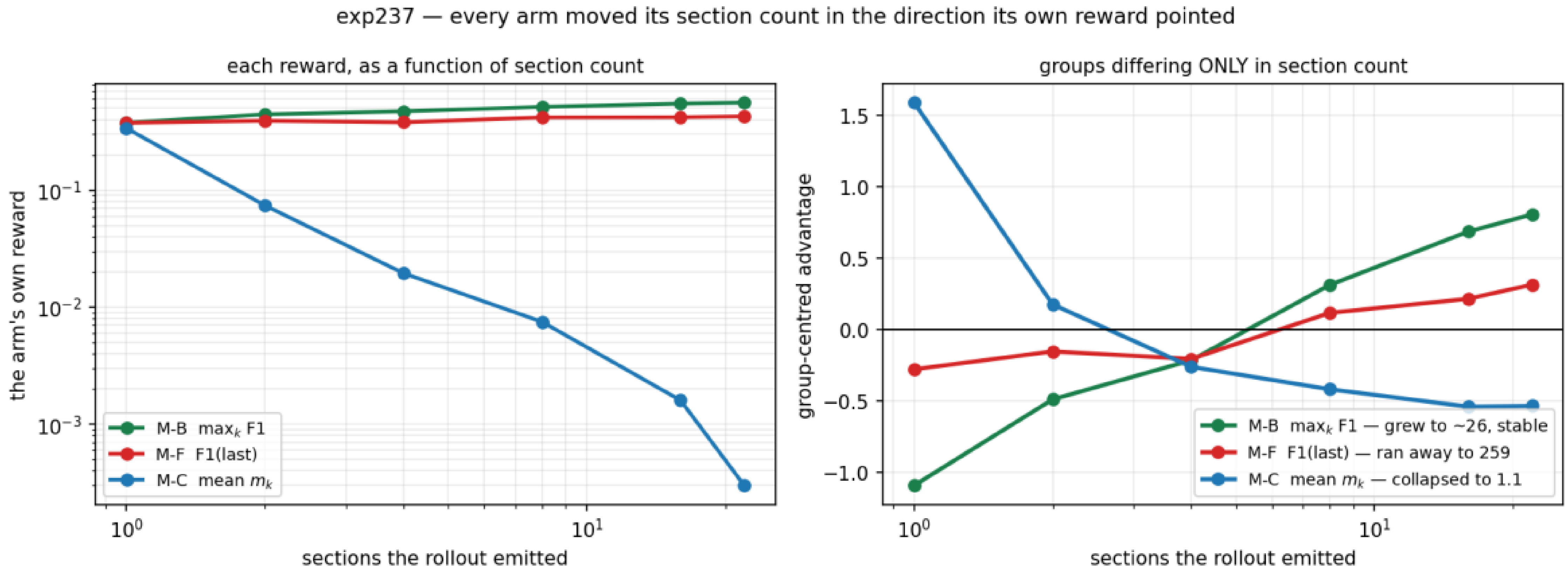
Arm M-C's reward before training: 55% of sections change no vote, and average -0.062 once centred. M-F has no such atom and collapsed the same -- so this is not the cause. See RESULTS.md.




```
$ plot_reward_shape.py --data data --out plots
```

reward_vs_section_count.png

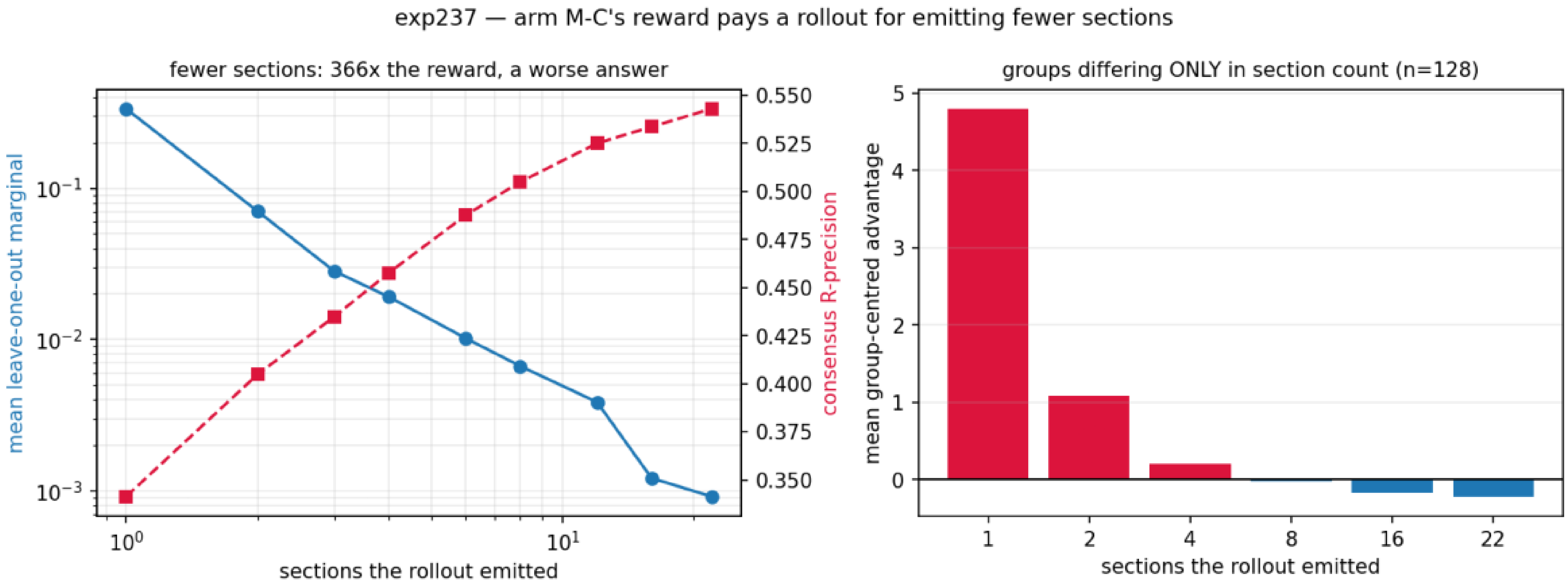
Each reward's gradient in the section-count direction predicts what that arm did: M-C falls (collapsed to 1.1), M-B rises (grew to ~26), M-F is nearly flat (unconstrained, ran to 259).



$ mkfig.py

section_count_incentive.png

m_k is not scale-free in the section count: at 1 section it is 366x its value at 22, while the rollout's own consensus falls 0.543 -> 0.341. Centring does not remove it -- the shorter rollout is the one above its group's mean.



$ plot_reward_shape.py --data data --out plots